ToricGT: Compact Graph-Token Reasoning with Tropical Ring Attention, Toric Diagnostics, and Random-Order Compression

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Abstract

We present ToricGT, a compact graph-to-graph Transformer family for algebraic, mathematical, morphological, and parameter-constrained language-model reasoning. The research model tokenizes nodes and edges as a shared sequence, uses exact blockwise ring evaluation for tropical max-plus attention, adds finite noncommutative-torus phase channels, and uses permutation-equivariant Soft-MoE routing in upper graph blocks. The paper contributes a finite theory and an implementation-facing training contract. We prove graph-token equivariance, exactness of tropical ring attention, stability of active faces under perturbation, and simulation of bounded semiring dynamic programs. Motivated by recent toric geometry of ReLU networks, we further interpret tropical head decisions as active faces of Newton polytopes and normal fans, giving compact diagnostics for margins, bends, and future toric auxiliary losses. A separate OpenAI Parameter-Golf track projects byte chunks to random-order graph positions and trains a dense recurrent micro language model with tied embeddings, toric position phases, tropical-ring attention, compact noncommutative-toric memory, strict-prefix GFlowNet graph-of-thought actions, BigramHash/CaseOps byte features, SmearGate confidence, stripped multi-token auxiliary heads, score-first test-time bias adaptation, and bit-packed 6-bit row-quantized export. The result is a small, auditable model design whose reasoning behavior is inspected through equivariance errors, active faces, toric moments, artifact bytes, BPB, and GFlowNet diversity rather than scalar accuracy alone.

1 Introduction

Many reasoning tasks are naturally graph-to-graph transformations: proof steps depend on earlier equations, graph algorithms update node and edge states, Hebrew morphology links surface forms to roots and templates, and algebraic words reduce through local rewrite relations. ToricGT targets this regime with three design constraints. First, relabeling a graph should relabel node and edge outputs, not change the computation. Second, dynamic-programming-like reasoning should be visible through max-plus active faces and margins. Third, compactness matters: the same ideas should support a small research model and a Parameter-Golf-style byte model under a strict artifact budget.

The main novelty is the integration of algebraic structure with trainable graph-token Transformers. Tropical attention implements max-plus gates and exposes provenance. Ring evaluation changes the schedule but not the function. Finite Weyl pairs encode projective toric phase memory. Soft-MoE gives typed differentiable routing in the research graph model. For Parameter Golf, the same graph-token idea is compressed into random-order autoregressive decoding: each byte position becomes a graph node scored in a content-independent random order. Finally, toric geometry supplies a new diagnostic layer: active candidates in tropical heads form exposed faces of Newton polytopes, and changes across cell walls behave like bend or Cartier-divisor intersection measurements.

2 Model

A ToricGT instance is

$$F(G) = D \circ \Phi_L \circ \cdots \circ \Phi_1 \circ T(G), \tag{1}$$

where T maps a typed attributed graph to node and edge tokens, Φ_ℓ is a shared token-equivariant block, and D is a shared node/edge/graph decoder. A graph is

$$G = (V, E, s, t, \tau_V, \tau_E, x, e, m_V, m_E), \tag{2}$$

with source and target maps for edge tokens, finite node and edge types, bounded attributes, and padding masks.

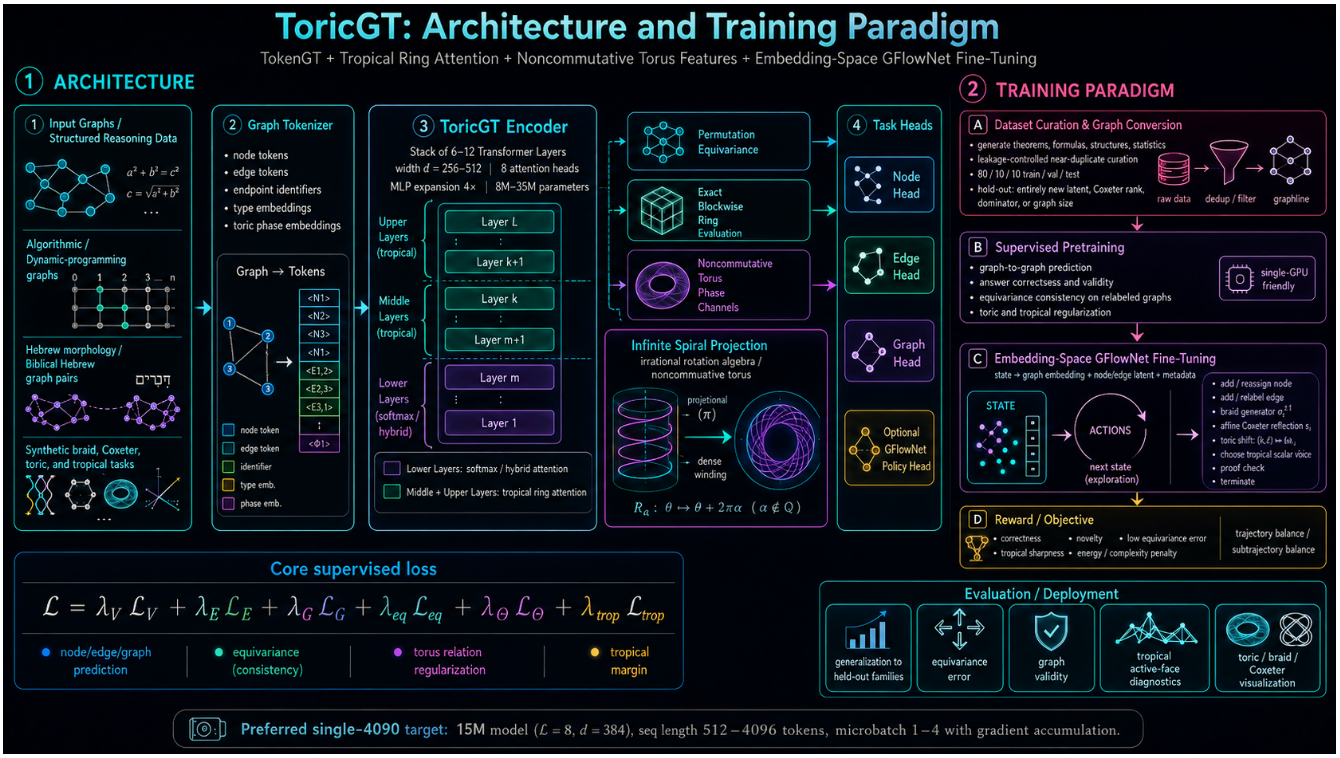

Figure 1: ToricGT combines graph tokenization, tropical ring attention, noncommutative-torus phase channels, Soft-MoE routing, and embedding-space GFlowNet fine-tuning.

Tropical ring attention. For score matrix S and values V, a tropical head computes

$$Y_{ic} = \max_j (S_{ij} + V_{jc}), \qquad A_{ic} = \arg\max_j (S_{ij} + V_{jc}). \tag{3}$$

Partitioning keys into blocks $B_1, \ldots, B_R$ gives

$$\max_j (S_{ij} + V_{jc}) = \max_r \max_{j \in B_r} (S_{ij} + V_{jc}), \tag{4}$$

so ring evaluation is exactly equal to full tropical attention in exact arithmetic.

Soft-MoE routing. For hidden states $X \in \mathbb{R}^{B \times L \times d}$, learned slot vectors $\Phi \in \mathbb{R}^{d \times E \times S}$, and validity mask M, define

$$A_{bies} = g \left\langle x_{bi} / \|x_{bi}\|, \phi_{es} / \|\phi_{es}\| \right\rangle, \tag{5}$$

$$D_{bies} = \frac{\exp A_{bies} M_{bi}}{\sum_j \exp A_{bjes} M_{bj} + \epsilon}, \tag{6}$$

$$C_{bies} = \frac{\exp A_{bies}}{\sum_{e',s'} \exp A_{bie's'} + \epsilon}, \tag{7}$$

$$Z_{bes} = \sum_i D_{bies} X_{bi}, \qquad Y_{bi} = \sum_{e,s} C_{bies} f_e(Z_{bes}). \tag{8}$$

We use $E \geq 4$ experts by default in the graph model. The Parameter-Golf micro-LM is dense by default because a full expert bank is usually a poor bytes-per-quality tradeoff under a 16 MB artifact cap; prefix-causal or low-rank Soft-MoE remains an ablation after the dense baseline is stable.

3 Theory

Theorem 1 (Graph equivariance). *If tokenization, masks, attention projections, tropical heads, Soft-MoE routing, residuals, normalization, and decoders are shared over token rows and use only equivariant incidence/type*

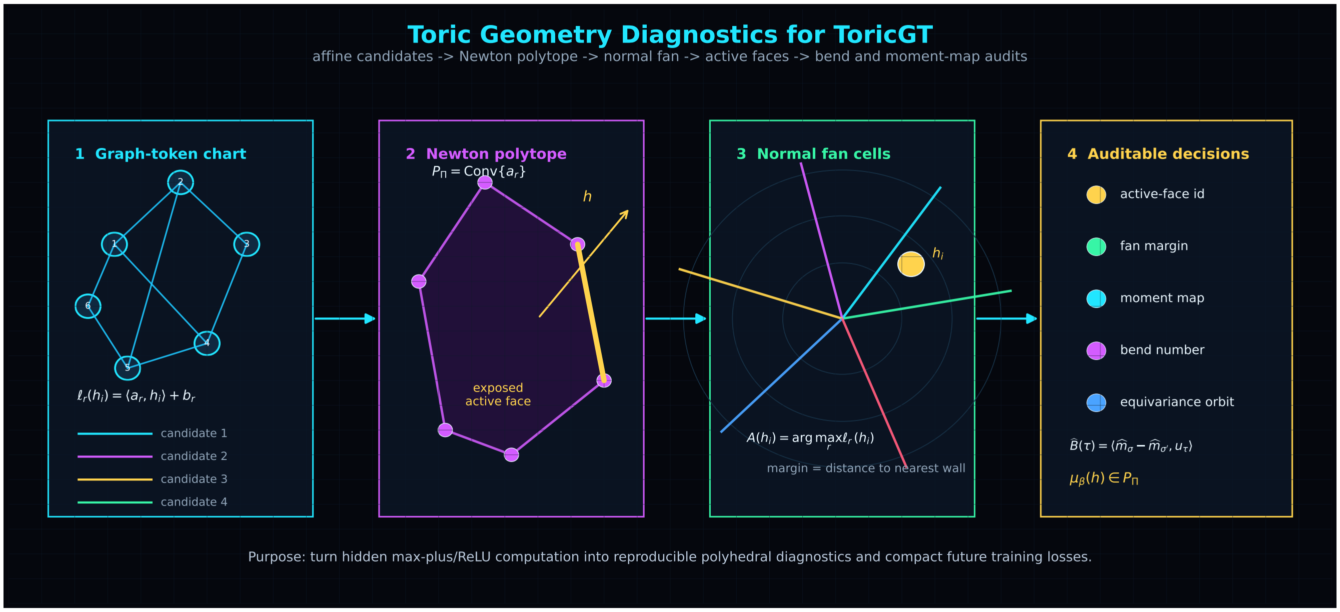

Figure 2: Toric diagnostics turn tropical candidate scores into Newton polytopes, normal-fan cells, active-face margins, and bend measurements.

information, then ToricGT is graph-to-graph equivariant.

Proof. Tokenization maps relabeled graphs to row-permuted token tables. Linear projections commute with row permutations. Softmax attention, tropical attention, and Soft-MoE dispatch/combination commute with the same row permutation because their normalizers sum over either all valid tokens or slots for a fixed token. Row-wise residual, normalization, and decoder maps commute with permutations. Composition preserves equivariance. □

Lemma 2 (Active-face stability). *If $\left\|S - \widehat{S}\right\|_\infty \le \epsilon_S$ and $\left\|V - \widehat{V}\right\|_\infty \le \epsilon_V$, then tropical outputs obey*

$$\left\|Y - \widehat{Y}\right\|_\infty \le \epsilon_S + \epsilon_V. \tag{9}$$

If the top-two margin is larger than $2(\epsilon_S + \epsilon_V)$, the active key is unchanged.

Proof. Each candidate score changes by at most $\epsilon_S + \epsilon_V$, and the maximum is 1-Lipschitz in ℓ_∞. The margin condition keeps the original maximizer strictly above all competitors after perturbation. □

Toric diagnostic theorem. Let a tropical probe be $\psi(h) = \max_r(\langle a_r, h\rangle + b_r)$ with rational slopes a_r. Its Newton polytope is $P = \mathrm{Conv}(a_1, \ldots, a_R)$. The active set

$$A(h) = \arg\max_r(\langle a_r, h\rangle + b_r) \tag{10}$$

is the exposed face of the lifted polytope $\mathrm{Conv}\{(a_r, b_r)\}_r$ under the linear functional $(a, b) \mapsto \langle a, h\rangle + b$. Hence tropical active-face diagnostics are cells of a normal fan. This imports the ReLU fan / Cartier-divisor viewpoint of Fu [1] into ToricGT tropical subcomputations, complementing earlier tropical rational interpretations of ReLU networks [2].

Finite noncommutative-toric memory. Let θ, β be rationally independent. The competition adapter stores a small memory bank indexed by

$$m_s = (\sin 2\pi s\theta, \cos 2\pi s\theta, \sin 2\pi s\beta, \cos 2\pi s\beta, \sin(2\pi s\theta + 2\pi s\beta + 2\pi s^2\theta), \cos(2\pi s(\theta - \beta))). \tag{11}$$

The quadratic phase term is a finite cocycle surrogate: it gives a compact, byte-accounted way to expose clock/shift-like projective phase structure without claiming a faithful finite representation of an irrational rotation algebra. Queries attend to these slots by cosine attention and add a small residual before logits.

Proposition 3 (Ergodic phase coverage)**.** *If $1, \theta, \beta$ are rationally independent, the sequence $(s\theta, s\beta) \bmod 1$ is equidistributed on $\mathbb{T}^2$. Consequently, finite prefixes of the toric memory supply low-discrepancy phase probes as the number of slots grows.*

Proof. This is Kronecker-Weyl equidistribution. For any nonzero $k \in \mathbb{Z}^2$, rational independence gives $k_1\theta + k_2\beta \notin \mathbb{Z}$, so the exponential averages $n^{-1}\sum_{s<n} \exp(2\pi i s(k_1\theta + k_2\beta))$ vanish. Weyl's criterion implies equidistribution. □

Hidden-flow regularization. Embedding reasoning paths are treated as discrete flows $h_0, \dots, h_T$. Let $u_t = h_t - h_{t-1}$ and $\Delta u_t = u_t - u_{t-1}$. The competition branch adds

$$\begin{aligned} \mathcal{L}_{\text{flow}} &= \mathbb{E}_t \|\Delta u_t\|_2^2 + \nu \mathbb{E}_t \|u_t\|_2^2 \\ &\quad + 10^{-3} \operatorname{Var}_t \|h_t\|_2. \end{aligned} \tag{12}$$

This is not a fluid simulation. It is a Navier–Stokes-inspired discrete dissipation prior: high-frequency trajectory oscillations are penalized, but smooth transport through embedding space remains cheap. The regularizer improves the stability of long GFlowNet reasoning paths and provides kinetic-energy and viscous-dissipation diagnostics for the 3D trajectory and energy-landscape plots.

DEC conservative reasoning flow. The new local-topology layer adapts a sharper idea from conservative discrete exterior calculus for incompressible Navier–Stokes on simplicial meshes [19]. In DEC, incidence matrices discretize d, diagonal Hodge stars discretize metric duality, and conservative schemes are audited by mass, vorticity, and kinetic-energy behavior. ToricGT uses the same algebraic pattern on reasoning complexes rather than on physical meshes. For a reasoning window and radius ρ, let $A_\rho^{\to}$ be the directed adjacency and define the skew 1-form

$$u_\rho = A_\rho^{\to} - (A_\rho^{\to})^{\mathsf{T}}. \tag{13}$$

The DEC audit residual is

$$\begin{aligned} \mathcal{L}_{\text{DEC}} = \; & \|\delta u_\rho\|_2^2 + \lambda_\omega \|\omega_{\rho+} - \omega_\rho\|_2^2 + \lambda_E (E_{\rho+} - E_\rho)^2 \\ & + \lambda_\star \left(E_\rho^\star / E_\rho - 1\right)^2 + \lambda_\wedge \left\|u_\rho \wedge \omega_\rho - i_{u_\rho}\omega_\rho\right\|_F^2. \end{aligned} \tag{14}$$

Here δ is the row-sum divergence proxy, ω_ρ is node circulation, E_ρ is edge-flow kinetic energy, $E_\rho^\star$ is the core-radius Hodge-weighted energy, and the final term is a local wedge/interior-product consistency check. The term is small and auxiliary; its purpose is to keep long graph-of-thought trajectories conservative enough for inference-time scaling without suppressing useful directed cycle structure.

4 Training And Evaluation

The supervised objective combines node, edge, and graph losses with equivariance, torus, tropical-margin, Hebrew morphology, Soft-MoE, cache, and distillation terms:

$$\begin{aligned} \mathcal{L} = \; & \lambda_V \mathcal{L}_V + \lambda_E \mathcal{L}_E + \lambda_G \mathcal{L}_G + \lambda_{\text{eq}} \mathcal{L}_{\text{eq}} + \lambda_\Theta \mathcal{L}_\Theta + \lambda_{\text{trop}} \mathcal{L}_{\text{margin}} \\ & + \lambda_H \mathcal{L}_H + \lambda_{\text{moe}} \mathcal{L}_{\text{moe}} + \lambda_{\text{cache}} \mathcal{L}_{\text{PQ}} + \lambda_{\text{KD}} \mathcal{L}_{\text{distill}}. \end{aligned} \tag{15}$$

Embedding-space GFlowNet fine-tuning samples reasoning graphs with probability proportional to reward using trajectory balance. The reward combines correctness, novelty, low equivariance error, tropical margin, low toric commutator error, and complexity penalties.

The data mixture is graph-first: math reasoning, graph-of-thought traces, algorithmic tasks, synthetic rotation/tropical tasks, Hebrew morphology graphs, and Jewish Hebrew text sources are normalized into Parquet records with stable content hashes and graph JSON. Splits are clustered by near-duplicate text, graph sketches, algebraic normal forms, root families, and source references.

Metric	Failure mode exposed
Equivariance error	absolute-position leakage and edge-order bugs
Active-face margin	unstable dynamic-programming decisions
Toric bend consistency	incoherent polyhedral reasoning changes
DEC conservation residual	noisy or nonconservative hidden reasoning flow
Soft-MoE expert mass	expert collapse or over-routing
GFlowNet diversity	mode collapse and reward hacking
Parameter-Golf BPB/bytes	compactness and scoring compliance

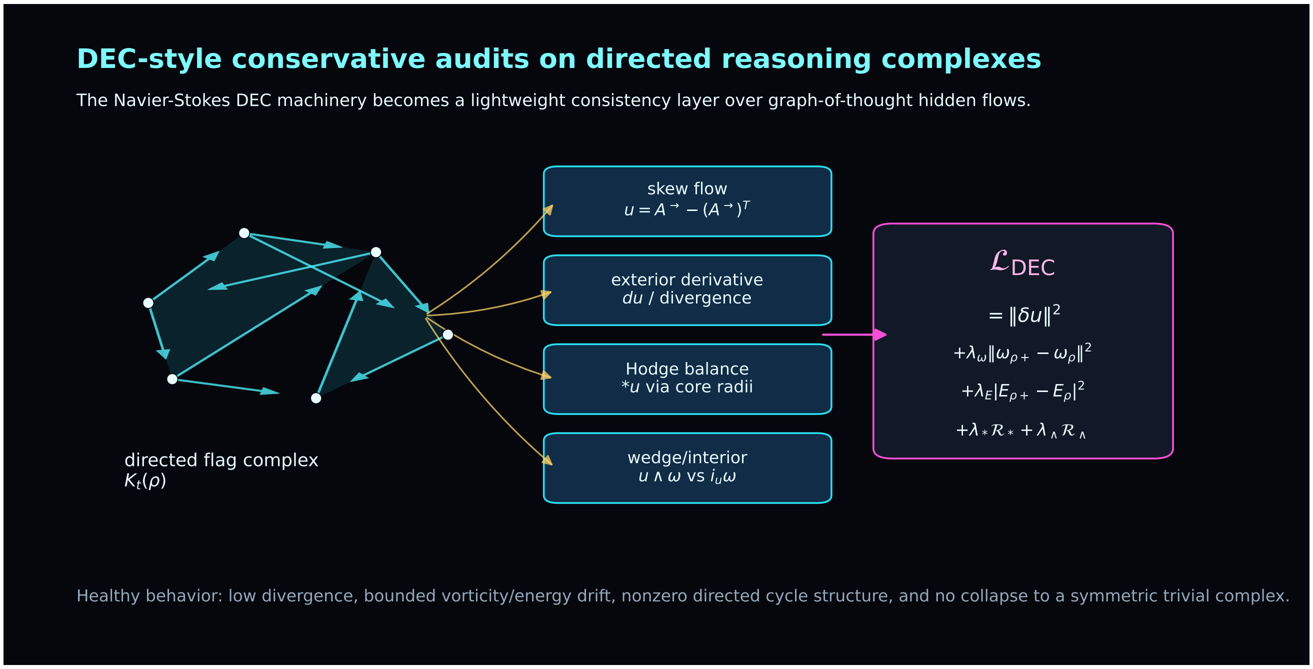

Figure 3: DEC-inspired conservative audits over directed reasoning complexes. The directed graph-of-thought adjacency is treated as a discrete 1-form; divergence, vorticity drift, Hodge balance, and wedge/interior consistency become lightweight diagnostics and low-weight training signals.

5 Parameter-Golf Track

The contest-facing model is not the full graph encoder. It is a dense random-order byte model with tied embeddings, recurrent block reuse, toric target-position phases, lower softmax attention, upper tropical-ring attention, compact prefix-visible GFlowNet action sampling, noncommutative-toric memory slots, and byte-accounted export. The artifact budget is

$$B_{\mathrm{art}} = B_{\mathrm{code}} + B_{\mathrm{weights}} + B_{\mathrm{tokenizer}} + B_{\mathrm{metadata}} \le 16{,}000{,}000. \tag{16}$$

For a byte chunk $b = (b_1, \ldots, b_T)$, sample a content-independent order $\pi \in S_T$ from public metadata and factor

$$q_\theta(b \mid \pi) = \prod_{k=1}^{T} q_\theta(b_{\pi_k} \mid b_{\pi_{<k}}, \pi_{\le k}). \tag{17}$$

BPB is computed prequentially: predict a normalized next-byte distribution, score the observed byte, then update the state. Because π is independent of unscored byte values and row k never contains b_{π_k}, random-order decoding is score-before-update valid while exposing more graph-like order diversity than ordinary left-to-right decoding. A byte-safe GFlowNet policy samples latent graph-of-thought action embeddings from the prefix-visible hidden state, and validation can average multiple action samples. Dense recurrence is the default because it adds computation without adding stored expert weights; Soft-MoE adapters are retained only as byte-accounted ablations. Curated `graph_json` records are projected into compact node/edge byte traces so graph structure is visible to this adapter.

The current competition branch adds several high-impact modifications inspired by OpenAI's public Parameter-Golf retrospective while excluding JEPA. BigramHash gives a tiny strict-prefix n-gram memory; CaseOps adds byte-class operator features; SmearGate learns input-dependent confidence; stripped multi-token heads improve hidden states but are omitted from the artifact; contrastive hidden-state regularization replaces JEPA-style prediction; coprime row striding changes data order without changing data; score-first output-bias adaptation updates only after a token has been scored; and bit-packed 6-bit row quantization with LZMA reduces serialized weight cost. Domain tags emphasize math, code, graph, Hebrew, biomedical, biochemical, biophysical, and toric records without adding evaluator dependencies. The local mirror of `AmelieSchreiber/toricgt-curated-splits` is used in the competition-safe split: train shards feed supervised BPB optimization, validation shards feed score-first validation, and test shards are reserved for held-out GFlowNet test-time scaling. The byte loader packs multiple source rows into one chunk; after warmup the training iterator switches to larger technical records, and a 2048-byte packed-context config can resume from a 1024-byte checkpoint by resizing only the learned position table. This exploits the tropical-ring upper layers for longer composite reasoning trajectories while preserving the strict prequential scoring contract.

The `oai` branch separates two ingredients. GraphCG learns steerable latent directions by contrasting edits along learned factors [3]; ToricGT uses the same idea to learn a small concept chart $B = [d_1, \ldots, d_r]$. The analogical-reasoning paper argues that analogy emerges when relational structures align in embedding space and a functor-like map is applied by the Transformer [4]. ToricGT composes these ideas: relation arrows and graph-of-thought states are first expressed in the GraphCG chart, while analogies are modeled as filtered topological functors. Let h_t be a prefix-visible hidden state, $\xi_t = B^\top h_t$, and let $a_t = (\xi_{t+s} - \xi_t)/(\|\xi_{t+s} - \xi_t\| + \epsilon)$ be a normalized chart relation arrow. Repeated coarse byte-relation classes are encouraged to share the same chart directions and to satisfy parallelogram closures. For scale robustness, each relation class builds a nested soft Vietoris–Rips filtration

$$A_\rho(i,j) = \sigma((\rho - \|a_i - a_j\| / \operatorname{med}\|a_p - a_q\|)/\tau), \qquad \rho_1 < \cdots < \rho_m, \tag{18}$$

and penalizes inclusion violations, chain-map commutators, and soft triangle failures. Directionality is supplied by an antisymmetric toric form

$$\Omega_{ij} = \langle a_i^{(1)}, a_j^{(2)} \rangle - \langle a_i^{(2)}, a_j^{(1)} \rangle, \qquad A_\rho^{\rightarrow}(i,j) = \sigma((\rho - d_{ij} + \gamma\Omega_{ij})/\tau), \tag{19}$$

so $i \to j$ and $j \to i$ may differ. In the hard-threshold limit these adjacencies define nested Vietoris–Rips or directed flag complexes

$$K_{\rho_1} \subseteq \cdots \subseteq K_{\rho_m}, \qquad H_p(K_{\rho_1}) \to \cdots \to H_p(K_{\rho_m}), \tag{20}$$

so the intended object is a small persistence module over hidden-state analogy arrows [15, 16, 17]. Consecutive reasoning windows also carry soft transports P_s that push source complexes to target complexes:

$$\widehat{A}_\rho^{s+1} = P_s^\top A_\rho^s P_s, \qquad \widehat{A}_\rho^{\rightarrow,s+1} = P_s^\top A_\rho^{\rightarrow,s} P_s. \tag{21}$$

Thus the maps are functor-like, but with additional topological structure: they should preserve filtration inclusions, approximately commute with chain and boundary maps, and induce low-residual morphisms of persistence modules. The implementation uses differentiable persistence surrogates rather than exact barcode backpropagation. A radius-parametrized HDBSCAN gate further computes core distances c_i, mutual reachability $d_{\mathrm{mr}}(i,j) = \max\{c_i, c_j, d_{ij}\}$, and persistent density affinities across the same radii. Only high-stability neighbors receive pullback weight; outlier arrows mostly stop contributing. The finite noncommutative toric memory supplies pseudo-periodic phase coordinates $(e^{2\pi i\theta t}, e^{2\pi i\beta t})$ and a quadratic cocycle surrogate; projected to a torus, these phases make recurrence, equidistribution, and directed skew visible along reasoning trajectories. Periodic analyses render the toric phase path, local simplicial edges, and soft analogical maps in 3D, together with directed filtration curves, HDBSCAN heatmaps, Ramachandran-style phase plots, energy landscapes, and reasoning/K/BPB simplices.

Operationally, the first serious BPB target is 1.20–1.22, a strong run is 1.16–1.19, an excellent run is 1.13–1.15, and ≤ 1.12 is exceptional under OpenAI's public recap. Claims below 1.10 BPB should trigger strict leakage, tokenizer, and score-before-update audits.

Kolmogorov-style reasoning diagnostics. The competition score is BPB, but BPB alone does not distinguish shallow byte compression from robust reusable reasoning. ToricGT therefore logs estimator-tagged proxies for conditional Kolmogorov complexity. For compressor c,

$$\widehat{K}_c(s) = |c(s)|, \qquad \widehat{K}_c(y \mid x) = \max\{0, \widehat{K}_c(x \,\|\, y) - \widehat{K}_c(x)\}, \tag{22}$$

and for two trajectories

$$\mathrm{NCD}_c(a,b) = \frac{\widehat{K}_c(a \,\|\, b) - \min(\widehat{K}_c(a), \widehat{K}_c(b))}{\max(\widehat{K}_c(a), \widehat{K}_c(b))}. \tag{23}$$

These are not claims to compute true Kolmogorov complexity. They are auditable lower-tech probes for whether successful graph-of-thought paths are compact conditional programs, whether random orders induce stable programs, and whether GFlowNet terminal diversity is genuine rather than cosmetic. The implementation logs conditional target complexity, order-program complexity, prediction-target NCD, graph-MDL summaries, and GFlowNet action-trace complexity to W&B. All such metrics are diagnostic by default; the BPB objective remains unchanged unless a future ablation explicitly gives a complexity reward nonzero weight.

The random-order model uses a structured helper set rather than a single string. At step k, let

$$H_k = \{b_{\pi_{<k}}, \pi, \mathrm{chunk}(b), a_{<k}, G_k, T_k\}, \tag{24}$$

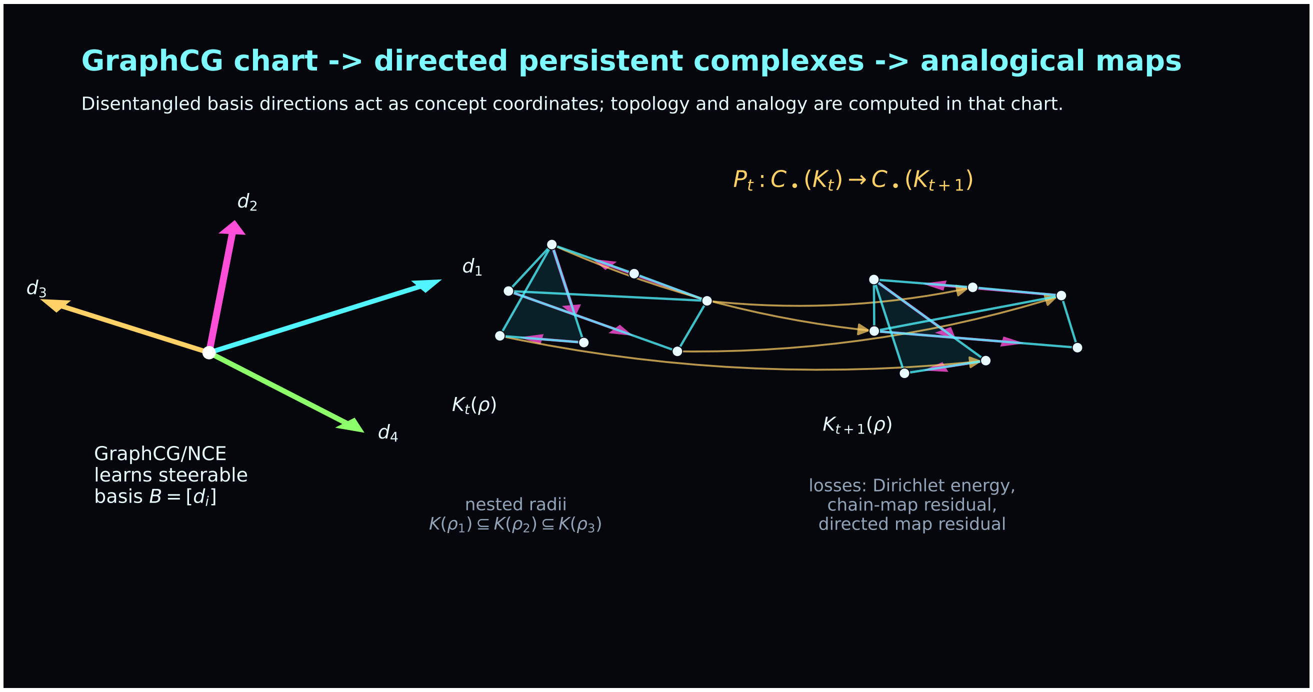

Figure 4: GraphCG supplies a steerable concept chart; ToricGT builds directed persistent complexes in that chart and learns soft functor-like maps that also preserve filtered topological structure.

where $b_{\pi_{<k}}$ is the strict prefix, π is the public random-order tree program, $\mathrm{chunk}(b)$ is the graph-projected byte chunk, $a_{<k}$ are prefix-visible GFlowNet actions, and G_k, T_k are optional serialized graph or tree helpers. The implemented proxy is

$$\widehat{K}_c(y_k \mid H_k) = \min_{h \in H_k \cup \{\varnothing\}} \widehat{K}_c(y_k \mid h). \tag{25}$$

Analogical transfer is measured by adding a source source-target helper pair:

$$\Delta K_c^{\mathrm{ana}} = \widehat{K}_c(y_t \mid H_t, x_s, y_s, G_s, T_s) - \widehat{K}_c(y_t \mid H_t). \tag{26}$$

Negative $\Delta K_c^{\mathrm{ana}}$ means the analogy shortened the target program. Prediction-side rewards are multiplied by a correctness gate, so a short wrong byte sequence does not count as a complexity improvement. The suite also logs a symmetry-of-information residual

$$\left| (\widehat{K}_c(x) - \widehat{K}_c(x \mid y)) - (\widehat{K}_c(y) - \widehat{K}_c(y \mid x)) \right| \tag{27}$$

to catch unreliable compressor asymmetries.

Reasoning simplex diagnostics. For evaluation, ToricGT projects compute-quality tradeoffs into small simplices. Let r be a normalized reasoning-budget score from trajectory length, recurrent passes, GFlowNet samples, and wall time; let k be a compressor-based complexity score; and let p be a low-BPB score. The triangle point is

$$z_\triangle = w_r v_r + w_k v_k + w_p v_p, \qquad (w_r, w_k, w_p) = \frac{(\epsilon + r, \epsilon + k, \epsilon + p)}{3\epsilon + r + k + p}. \tag{28}$$

Sparse budget evaluations are interpolated inside the triangle by a radial basis kernel, producing a dark-to-light blue heatmap whose outward flow shows how additional reasoning time moves the model toward complexity, compression, or reasoning vertices. A tetrahedral extension adds a fourth coordinate: either GFlowNet diversity or hidden-trajectory minimum-spanning-tree efficiency. The MST is computed over prefix hidden states with Euclidean edge weights and gives a compactness proxy for the reasoning path; high quality with low MST length indicates organized reasoning rather than noisy wandering.

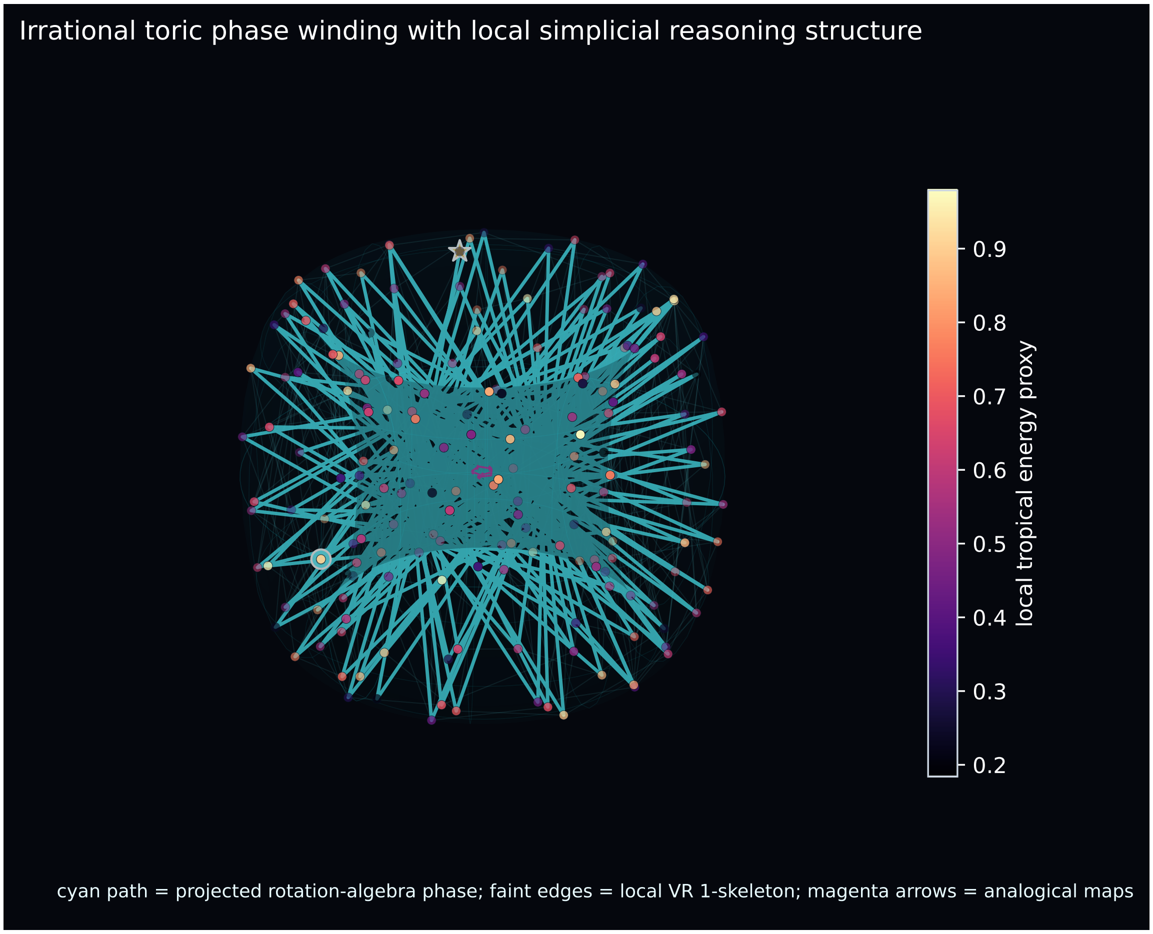

Figure 5: Irrational toric phase projection of a reasoning trajectory. The torus path shows pseudo-periodic phase coverage; faint local edges are Vietoris–Rips 1-simplices; magenta arrows represent window-to-window analogical maps.

6 Next Iteration

The next ToricGT iteration uses toric geometry as a training signal while keeping the competition artifact dense and small. First, freeze checkpoints and build empirical toric shadows: occupied fan cells, fitted slopes, active-face margins, and bend magnitudes. Second, add synthetic toric tasks: Newton-polytope active-face prediction, Cartier-divisor bend prediction, toric ideal relation checks, and ReLU fan realization tests. Third, attach training-only low-rank quantized toric probes and optimize

$$\mathcal{L}_{\text{toric}} = \mathcal{L}_{\text{base}} + \lambda_{\text{fan}}\mathcal{L}_{\text{fan}} + \lambda_{\text{bend}}\mathcal{L}_{\text{bend}} + \lambda_{\text{binom}}\mathcal{L}_{\text{binom}} + \lambda_{\text{moment}}\,\|\widehat{\mu} - \mu^{\star}\|_2^2 + \lambda_{\text{cox}}\mathcal{L}_{\text{cox}} + \lambda_{\text{br}}\mathcal{L}_{\text{br}} + \lambda_{\text{leaf}}\mathcal{L}_{\text{leaf}}. \tag{29}$$

The current `oai` implementation instantiates these terms as a 6-bit fake-quantized probe that is excluded from artifact export. Its metrics report active-face cross entropy, top-two margins, moment-map residuals, bend magnitude, toric binomial residual, affine-wall distance, Coxeter reflection loss, braid loss, and phase-leaf residual. The analysis suite renders a toric-shadow audit in which branch fan coverage, margins, bends, and leaf residuals can be compared directly against BPB and graph-of-thought quality.

Affine toric charts and Koszul persistence. The next diagnostic step is to turn each active fan cell into a small affine scheme. If $\sigma \subset N_{\mathbb{R}}$ is a rational cone selected by a tropical active face, then the local toric chart is

$$U_\sigma = \operatorname{Spec} k[\sigma^\vee \cap M], \qquad R_\sigma = k[\sigma^\vee \cap M]. \tag{30}$$

Reasoning trajectories filtered by time, radius, density, margin, and phase are then finite multigraded modules over $k[x_1, \ldots, x_n]$, or over the chart ring R_σ when toric structure is active. This matches the commutative-algebra view of multiparameter persistence as finitely generated graded modules [18]. Given local parameters $a_1, \dots, a_r$, attach the Koszul complex

$$K_q(a; M) = M \otimes \bigwedge^q k^r, \qquad H_q(K(a; M)) \cong \mathrm{Tor}_q^R(R/(a), M). \tag{31}$$

Nonzero higher homology, large Fitting-minor residuals, failed varieties-of-complexes equations $d_i d_{i+1} = 0$, unstable multigraded Betti mass, or Buchsbaum–Eisenbud complementary-minor multiplier mismatches indicate that the model's claimed toric/persistence coordinates are not coherent local parameters for the observed reasoning module. These quantities are audit-first: they must pass exact small-window checks and ablations before becoming training losses.

Affine Coxeter and braid structure. The toric route to affine Coxeter structure is through lattices and chambers. A torus T has character and cocharacter lattices $M = \mathrm{Hom}(T, \mathbb{G}_m)$ and $N = \mathrm{Hom}(\mathbb{G}_m, T)$. Once a root datum is attached, as for a reductive group with maximal torus T, roots $\alpha \in M$ cut $N_{\mathbb{R}}$ by hyperplanes $\alpha(x) = 0$; the reflections in these walls generate the Weyl group, and the translated walls $\alpha(x) = k$ generate the affine Weyl/Coxeter group $W_{\mathrm{aff}} = Q^\vee \rtimes W$ [12]. Thus toric fans and Newton normal fans give the polyhedral side of the same chamber geometry that appears in affine Coxeter systems. Artin braid actions then arise as ordered wall-crossing or monodromy actions around the complexified hyperplane complement. In the next iteration, braid/Coxeter graph tasks should therefore be tied to toric fan tasks: actions $\sigma_i^{\pm 1}$ and affine reflections $s_{\alpha,k}$ become controlled graph-of-thought edits, while tropical active-face margins determine whether a wall crossing is stable or ambiguous.

Noncommutative phase foliations. The rotation algebra relation $VU = e^{2\pi i\theta} UV$ supplies a projective phase over the ordinary torus shadow. Forgetting operator order maps a word to its commutative degree $(a, b) \in \mathbb{Z}^2$, while retaining the cocycle exponent records the extra phase accumulated by crossings. The audit projection

$$k \longmapsto \left(e^{2\pi i k\theta}, e^{2\pi i k\beta}\right) \in \mathbb{T}^2 \tag{32}$$

with irrationally related θ, β gives a Kronecker foliation of the commutative torus by dense phase leaves; rational approximants give closed leaves. ToricGT should use these leaves as phase-consistency diagnostics for reasoning trajectories: hidden graph-of-thought paths should move smoothly along projected leaves unless a verified algebraic or tropical wall crossing occurs. This is only a finite audit shadow of the rotation C^*-algebra, not a faithful finite representation, but it gives a concrete way to test whether noncommutative toric memory is organizing latent search rather than adding decorative sinusoidal features [10].

The same finite phase path admits a Slepian audit. For a window of length N and normalized half-bandwidth W, the discrete time-band limiting operator

$$K_W[m, n] = \begin{cases} 2W, & m = n, \\ \dfrac{\sin(2\pi W(m-n))}{\pi(m-n)}, & m \neq n \end{cases} \tag{33}$$

has discrete prolate spheroidal eigenvectors. Projecting a toric phase signal $s_k = s(e^{2\pi i k\theta}, e^{2\pi i k\beta})$ onto its leading eigenvectors gives concentration, leakage, and mode-entropy diagnostics. High concentration means the projected noncommutative phase memory is coherent and time-band localized; high leakage means the hidden trajectory has diffuse phase drift. The implementation logs these metrics and renders a toric Slepian panel, but keeps them audit-only until ablations show that optimizing them improves BPB or reasoning robustness.

Finally, make inference-time scaling geometric: continuous GFlowNets can move inside normal cones, along walls, across moment polytopes, and along projected toric leaves while discrete actions edit graph-of-thought states, apply braid generators, or apply affine Coxeter reflections. Continuous GFlowNet theory supplies the density-correct trajectory-balance objective for these hybrid paths [9]. This keeps the architecture small while giving search a structured latent space.

7 Conclusion

ToricGT is a finite, auditable model family. Its exact claims are graph-token equivariance, ring-schedule exactness, max-plus semiring simulation on bounded domains, projective toric phase consistency, random-order score validity for byte modeling, and perturbation bounds for compressed caches. Its practical claim is that reasoning models should expose the geometry of their decisions. Tropical active faces, Newton polytopes, toric bends, Koszul residuals, artifact bytes, BPB, expert utilization, and GFlowNet reward landscapes make small-model reasoning easier to test, compress, and improve. The guardrail for skeptical readers is simple: no advanced geometric diagnostic changes training unless it has a logged metric, an ablation, and a plot.

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